Building an AI Engine Optimization (AEO) System for Measuring How Well LLMs Answer Snowflake Developer Questions

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1 Snowflake, Developer Relations

Summary

AI coding assistants are now part of the Snowflake developer workflow, but there is no systematic way to measure whether those assistants give developers correct, current answers. We built an AEO benchmark that evaluates AI answer quality across 128 Snowflake developer questions spanning 32 product categories. Using a 2^4 factorial experiment design, we tested 16 combinations of four augmentation factors (domain prompt, citation instruction, agentic tools, self-critique) to isolate what actually improves answer quality. The best configuration (citation + agentic tools, no domain prompt) scored 82.3%, a 29.1 percentage-point improvement over the bare LLM baseline of 53.2%. Agentic tool access was the dominant factor (+10.9pp average), while self-critique was consistently counterproductive (−2.7pp average). These findings directly inform how Snowflake should configure its AI-powered developer tools.

1 Introduction

When a developer asks an AI assistant "How do I create a Cortex Search Service?", the quality of the answer depends on more than the model's training data. It depends on whether the assistant can search current documentation, whether it has domain-specific prompting, and whether it applies post-generation review. But which of these augmentation strategies actually matter, and how much?

In early 2026, Vercel built an AI Engine Optimization (AEO) system to track how AI coding agents reference and recommend their products. Their system measures brand visibility: does the agent mention Vercel products? Our system asks a different question: does the agent give the *right* answer?

We designed a benchmark that goes beyond brand tracking to measure multi-dimensional answer quality (correctness, completeness, recency, citation, and recommendation) against expert-authored canonical answers. Rather than testing multiple competing agents, we tested multiple augmentation configurations on a single model (`claude-opus-4-6`) to isolate the effect of each deployment lever. The result is a data-backed framework for configuring AI developer tools on the Snowflake platform.

2 Methodology

2.1 Question Bank

We authored 128 questions across 32 Snowflake product categories, with exactly 4 questions per category. Categories span the full Snowflake developer surface including

Cortex AI Functions, Cortex Search, Cortex Agents, Cortex Code, Dynamic Tables, Snowpark, Streamlit in Snowflake, Apache Iceberg Tables, Snowflake ML, Snowpark Container Services, Native Apps Framework, Data Pipelines (Streams, Tasks, Snowpipe), Data Governance & Security, Snowflake Fundamentals & Architecture, dbt Projects on Snowflake, Semantic Views & Cortex Analyst, Database Change Management, Snowflake Postgres, and others. Each question falls into one of four test types: Explain (32), Implement (32), Debug (32), or Compare (32). For every question, we wrote a canonical answer grounded in official Snowflake documentation and defined up to five must-have factual elements (up to 640 total binary pass/fail checks).

Q#	Type	Question
Q1	Explain	What Cortex AI functions does Snowflake offer for text and image analytics, and how do they differ from each other?
Q7	Implement	Write the SQL to create a Cortex Search Service on a product catalog table, with filtering by category and a 1-hour target lag.
Q15	Debug	My Dynamic Table is doing full refreshes instead of incremental. How do I diagnose and fix this?
Q21	Compare	What is the difference between a UDF, UDTF, UDAF, and a vectorized UDF in Snowflake? When should I use each?

Table 1. Question Bank Excerpt. One representative question per test type (Explain, Implement, Debug, Compare), drawn from the 128-question bank spanning 32 Snowflake product categories.

2.2 Scoring Rubric

Each response was scored on five dimensions using a 0–10 scale:

- **Correctness**: factually accurate per current Snowflake docs
- **Completeness**: covers all key concepts and steps
- **Recency**: uses current syntax and feature names
- **Citation**: references or directs to Snowflake documentation
- **Recommendation**: recommends the Snowflake-native approach

Maximum score per question: 50 points (5 dimensions $\times$ 10). Maximum per run: 6,400 points (128 questions). Each question also has up to five must-have binary checks, producing a separate must-have (MH) pass rate.

2.3 Judge Panel

Every response was scored independently by three LLM judges: `openai-gpt-5.4`, `claude-opus-4-6`, and `llama4-maverick`. The final score for each response is the panel average. This design mitigates single-model scoring bias.

2.4 2^4 Factorial Experiment

We tested four binary augmentation factors in all 16 possible combinations:

Factor	OFF	ON
Domain Prompt	No system message	1,800-token Snowflake product knowledge primer
Citation	Raw question only	"Cite official Snowflake docs" appended to question
Agentic Tools	Single `CORTEX.COMPLETE` call (parametric only)	Cortex Code subagent with web search, doc search, skills
Self-Critique	Single-turn generation	Two-turn generate-then-revise

Non-agentic runs (8 of 16) used `SNOWFLAKE.CORTEX.COMPLETE('claude-opus-4-6', ...)` with the conversation format. Agentic runs (8 of 16) launched parallel Cortex Code subagents with full tool access. All 16 runs used `claude-opus-4-6` exclusively as the respondent model to avoid cross-model contamination.

Runs are numbered in Yates order: run = 1 + D + 2C + 4A + 8S, where D, C, A, S are 0 or 1. Each factor has a fixed bit weight, so any configuration is directly decodable from its run number.

Subagents were sandboxed to prevent access to canonical answers, scoring rubrics, or benchmark files.

3 Results

3.1 Overall Rankings

The 16 configurations produced scores ranging from 53.2% to 82.3%. Config abbreviations: D = Domain Prompt, C = Citation, A = Agentic, S = Self-Critique; Baseline = all factors OFF.

Config	Domain	Citation	Agentic	SC	Score	MH
C+A		✓	✓		**82.3%**	87.7%
D+C+A	✓	✓	✓		76.0%	82.7%
A			✓		74.4%	96.3%
D+C+A+S	✓	✓	✓	✓	73.5%	71.8%
C+A+S		✓	✓	✓	72.4%	69.0%
D+A	✓		✓		69.4%	86.0%
C		✓			67.7%	62.4%
C+S		✓		✓	67.2%	55.0%
D+C	✓	✓			66.1%	64.8%
A+S			✓	✓	66.1%	76.6%
D+C+S	✓	✓		✓	66.1%	57.8%
D+A+S	✓		✓	✓	65.4%	76.3%
D+S	✓			✓	58.4%	63.6%
D	✓				57.8%	66.1%
S				✓	56.1%	60.5%
Baseline					**53.2%**	62.7%

The engine split is clear: Cortex Code subagents (with tool access) averaged 72.4% score and 80.8% MH, compared to 61.6% score and 61.6% MH for single `CORTEX.COMPLETE` calls. The agentic/non-agentic divide is clean at this scale: the top 6 configurations are all agentic, while the bottom 10 are predominantly non-agentic. The 29.1pp score range (53.2% to 82.3%) is narrower than a preliminary 50-question pilot where the range was 37pp, reflecting that a broader question bank dampens configuration-specific variance and produces more stable rank ordering.

3.2 Main Effects

The factorial design lets us compute the average impact of turning each factor ON across all 8 paired comparisons:

Factor	Score Effect	MH Effect
Agentic Tools	**+10.9pp**	**+19.2pp**
Citation Instruction	+8.8pp	−4.6pp
Domain Prompt	−0.8pp	−0.1pp
Self-Critique	−2.7pp	−9.8pp

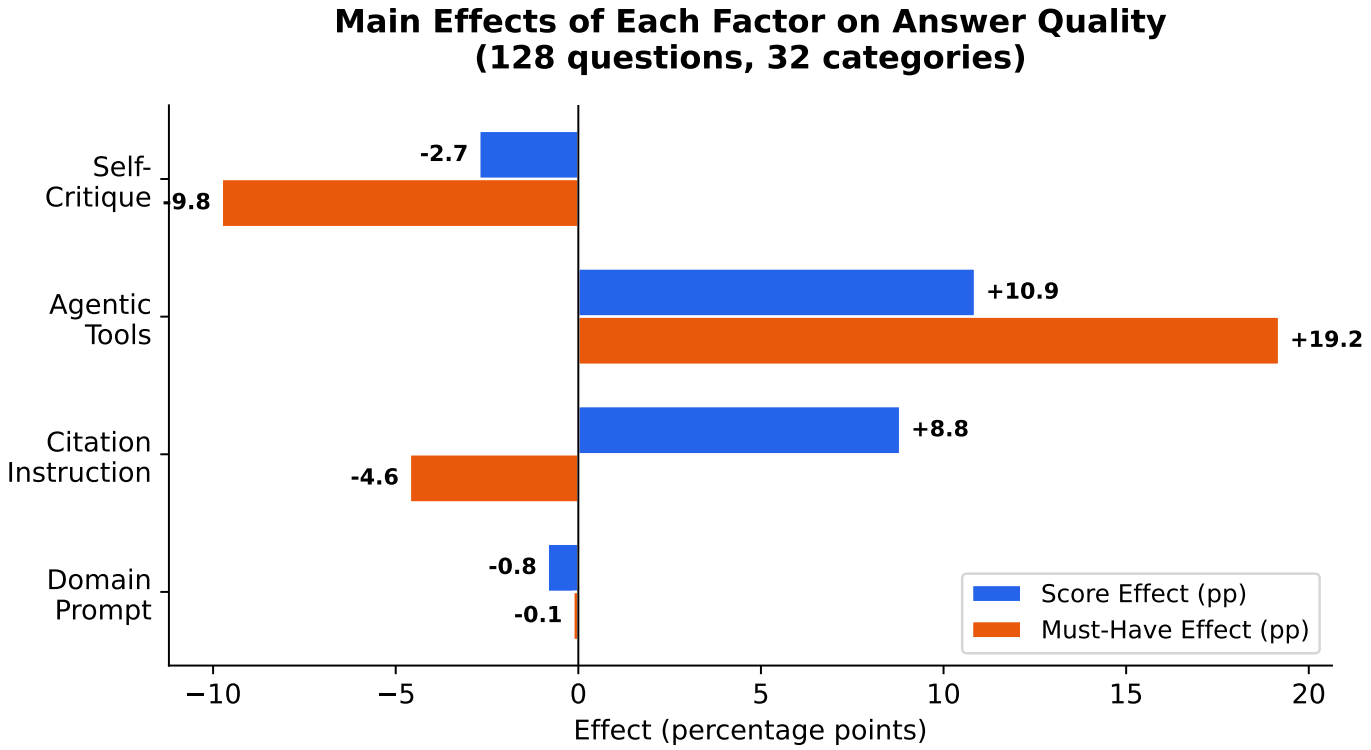

Figure 1. Main effects bar chart. Agentic tools dominate both score and must-have compliance. Self-Critique and Domain Prompt show negative effects on average.

Agentic tools are the dominant factor. They improve both score and must-have compliance in every single paired comparison. The ability to search current documentation and invoke specialized skills is more valuable than any prompting strategy.

Citation instruction helps score but hurts must-have compliance. The +8.8pp score effect comes largely from the Citation dimension itself (which rises sharply when citation instruction is active, especially when combined with agentic tools that can actually retrieve and link real documentation URLs). The −4.6pp MH effect suggests that instructing the model to cite sources sometimes causes it to pad responses with references at the expense of core factual coverage.

Domain prompt is net negative. Across all 8 paired comparisons the domain prompt shows a marginal −0.8pp average score effect. A 1,800-token primer cannot meaningfully cover 32 product categories; as the question bank grows, the primer's per-category coverage shrinks and its interference with retrieved information increases. In non-agentic conditions it provides a slight context benefit, but in agentic conditions the static system prompt actively constrains how the agent applies its retrieved content.

Self-critique is counterproductive on both metrics. It hurts score (−2.7pp) and dramatically hurts must-have pass rate (−9.8pp) across the board. The two-turn

"generate then revise" pattern causes the model to second-guess correct content, introduce hedging, and sometimes remove accurate details present in the first pass.

3.3 Category Performance

The table below compares the baseline (all factors OFF) against the best configuration (C+A: citation + agentic tools) across all 32 product categories:

Category	Baseline	Best (C+A)	Delta
Data Governance & Security	66.2%	90.8%	+24.6pp
dbt Projects on Snowflake	42.2%	90.3%	+48.1pp
Cortex Code	49.8%	88.8%	+39.0pp
Apache Iceberg Tables	69.3%	88.3%	+19.0pp
Cortex Search	59.3%	88.3%	+29.0pp
Cost Management	51.0%	87.5%	+36.5pp
Snowflake Fundamentals & Architecture	62.2%	87.5%	+25.3pp
Collaboration & Data Sharing	55.3%	86.8%	+31.5pp
Streamlit in Snowflake	61.3%	86.3%	+25.0pp
Snowpark	61.5%	86.2%	+24.7pp
Snowpark Container Services	66.0%	85.3%	+19.3pp
Cortex AI Function Studio	54.7%	84.7%	+30.0pp
Snowflake ML	53.7%	84.3%	+30.6pp
Database Security	43.0%	84.2%	+41.2pp
Openflow	26.7%	83.2%	+56.5pp
Cortex AI Functions	43.7%	83.0%	+39.3pp
Native Apps Framework	47.8%	82.8%	+35.0pp
Hybrid Tables	59.5%	82.8%	+23.3pp
SQL Performance & Optimization	59.5%	82.7%	+23.2pp
Data Pipelines (Streams, Tasks, Snowpipe)	54.5%	81.5%	+27.0pp
Snowpark Connect & Migration	51.3%	81.2%	+29.9pp
Snowflake Notebooks (Workspaces)	58.3%	80.8%	+22.5pp
AI Observability & Evaluation	46.3%	80.2%	+33.9pp
Data Clean Rooms	46.5%	79.3%	+32.8pp
Semantic Views & Cortex Analyst	48.8%	79.2%	+30.4pp
Cortex Agents	49.5%	78.5%	+29.0pp
Database Change Management (DCM)	46.2%	78.0%	+31.8pp
Data Loading (COPY, Snowpipe, Streaming)	55.6%	77.2%	+21.6pp
Snowsight	61.2%	76.8%	+15.6pp
Dynamic Tables	57.7%	75.0%	+17.3pp
Data Quality & Observability	54.3%	69.2%	+14.9pp
Snowflake Postgres	38.0%	61.8%	+23.8pp

Every category improved. The largest gains occurred in Openflow (+56.5pp), dbt Projects on Snowflake (+48.1pp), and Database Security (+41.2pp). Openflow also had the lowest baseline score of any category (26.7%), reflecting near-total absence of this feature in parametric model knowledge; agentic retrieval effectively filled that gap. At the other end, Snowflake Postgres remained the weakest category even under the best configuration (61.8%), driven by the operational specificity of its questions (connection management, diagnostic queries, storage integration) where even targeted retrieval did not reliably surface complete procedural answers. Data Quality & Observability also scored below 70% under C+A (69.2%), suggesting that newer platform capabilities with rapidly evolving documentation present a persistent challenge regardless of augmentation strategy.

Figure 2. Category Improvement: Baseline vs Best Configuration (128 questions, 32 categories). Per-category improvement from the baseline (all factors OFF) to the best configuration (C+A: citation + agentic tools). Categories sorted by delta. Every category improved, with Openflow gaining the most (+56.5pp).

3.4 Scoring Dimension Analysis

Citation is the dimension most sensitive to configuration. Without explicit citation instruction, models score near zero on the Citation dimension (1.1/10 in the baseline). With citation instruction plus agentic tools, it reaches 7.8/10. The other four dimensions are more stable, with agentic access providing consistent lifts across all of them. The table below shows per-dimension scores for the baseline and best run:

Config	**Correct.**	**Complete.**	**Recency**	**Citation**	**Recommend.**
Baseline	67.0%	58.6%	67.0%	11.3%	61.8%
C+A (Best)	86.3%	79.0%	85.7%	78.4%	82.0%
Delta	+19.3pp	+20.4pp	+18.7pp	**+67.1pp**	+20.2pp

3.5 Full Factorial Heatmap

The heatmap below shows all 16 runs against all 32 product categories, with rows grouped by whether the agentic tools factor is ON (upper half) or OFF (lower half). The color gradient makes the agentic divide immediately visible: the upper half is predominantly green (high scores), while the lower half shifts toward yellow and red.

	Agentic									Non-Agentic									
	Avg	C+A*	D+C+A	A	D+C+A+S	C+A+S	D+A	A+S	D+A+S	C	C+S	D+C	D+C+S	D+S	D	S	Baseline	Avg	Overall Avg
Data Governance & Security	**80.5**	91	89	77	87	83	74	67	75	80	72	77	75	64	64	64	66	**70.3**	**75.4**
dbt Projects on Snowflake	**74.2**	90	76	78	73	74	79	64	60	64	61	62	56	52	45	35	42	**52.3**	**63.3**
Cortex Code	**74.8**	89	79	83	83	81	60	63	61	64	65	66	55	54	52	60	50	**58.2**	**66.5**
Apache Iceberg Tables	**78.1**	88	80	76	84	84	72	71	70	78	76	75	74	69	70	65	69	**72.3**	**75.2**
Cortex Search	**79.3**	88	88	81	77	82	76	72	70	74	75	70	77	59	58	60	59	**66.5**	**72.9**
Cost Management	**70.3**	88	69	75	62	76	67	63	64	71	68	66	66	60	62	56	51	**62.7**	**66.5**
Snowflake Fundamentals & Arch.	**77.9**	88	83	76	83	79	74	73	69	79	73	72	71	54	67	53	62	**66.5**	**72.2**
Collaboration & Data Sharing	**74.3**	87	74	78	72	76	71	70	68	78	78	75	69	63	58	56	55	**66.4**	**70.4**
Streamlit in Snowflake	**76.9**	86	66	79	84	74	75	75	76	78	78	76	81	68	74	71	61	**73.4**	**75.2**
Snowpark	**77.6**	86	79	73	83	82	73	73	72	73	71	76	72	66	64	62	62	**68.3**	**72.9**
Snowpark Container Services	**73.3**	85	76	78	66	80	69	68	64	78	79	74	79	68	70	68	66	**72.8**	**73.0**
Cortex AI Function Studio	**74.7**	85	77	73	84	63	72	73	70	66	70	66	57	57	62	57	55	**61.2**	**67.9**
Snowflake ML	**74.4**	84	80	76	80	80	66	66	63	63	71	58	69	56	60	63	54	**61.9**	**68.1**
Database Security	**74.2**	84	79	77	73	75	67	72	66	71	70	68	72	61	62	50	43	**62.1**	**68.2**
Openflow	**69.7**	83	76	75	61	64	74	58	66	47	49	53	45	52	48	40	27	**45.0**	**57.3**
Cortex AI Functions	**68.3**	83	81	79	68	67	67	52	51	60	54	64	63	57	54	52	44	**55.9**	**62.1**
Hybrid Tables	**72.1**	83	70	66	74	79	63	67	76	74	76	73	74	62	67	61	60	**68.2**	**70.1**
Native Apps Framework	**69.6**	83	60	74	72	75	56	68	68	73	53	69	54	56	47	52	48	**56.3**	**63.0**
SQL Performance & Optimization	**72.0**	83	84	67	77	68	67	68	62	73	64	65	70	64	62	47	60	**63.2**	**67.6**
Data Pipelines (Streams/Tasks)	**72.7**	82	76	66	80	70	75	65	68	69	67	62	75	65	64	63	54	**65.0**	**68.8**
Snowpark Connect & Migration	**72.9**	81	78	75	73	70	70	70	66	56	59	50	57	52	51	46	51	**52.8**	**62.8**
Snowflake Notebooks	**73.2**	81	80	70	78	73	69	66	68	74	75	75	72	60	61	60	58	**67.0**	**70.1**
AI Observability & Evaluation	**64.7**	80	67	69	62	62	62	52	62	59	55	54	65	55	48	49	46	**53.8**	**59.3**
Data Clean Rooms	**69.8**	79	75	70	72	70	72	57	64	59	68	57	63	57	62	56	46	**58.5**	**64.1**
Semantic Views & Cortex Analyst	**69.5**	79	72	71	78	59	68	69	59	62	65	66	56	51	52	52	49	**56.6**	**63.0**
Cortex Agents	**66.1**	78	72	78	57	64	74	58	48	66	66	60	66	56	48	56	50	**58.5**	**62.3**
Database Change Management	**69.3**	78	71	76	63	70	68	63	66	62	64	66	56	55	52	56	46	**57.2**	**63.2**
Data Loading (COPY/Snowpipe)	**71.0**	77	81	74	68	66	68	68	66	68	69	66	72	57	55	57	56	**62.5**	**66.8**
Snowsight	**72.2**	77	80	70	64	76	73	69	68	71	73	76	68	59	61	66	61	**67.1**	**69.6**
Dynamic Tables	**71.9**	75	72	78	74	76	68	70	63	68	70	66	64	65	63	61	58	**64.3**	**68.1**
Data Quality & Observability	**68.6**	69	70	74	72	73	64	70	56	71	68	68	70	48	52	58	54	**61.2**	**64.9**
Snowflake Postgres	**64.2**	62	71	72	71	48	69	56	66	40	49	44	50	45	34	41	38	**42.7**	**53.5**

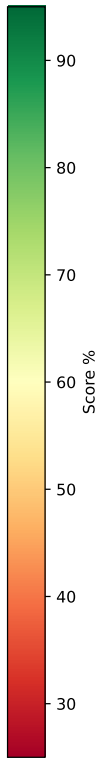

Figure 3. Category-level heatmap of all 16 factorial runs across 32 product categories. Rows are grouped by Agentic factor (upper = ON, lower = OFF) and sorted by average score within each group. The rightmost column shows the run average. Factor abbreviations: D = Domain Prompt, C = Citation, A = Agentic, S = Self-Critique.

4 Conclusion

Four actionable findings emerge from this benchmark:

1. **Deploy agentic tools, not bigger prompts.** Access to current documentation and specialized skills produces larger quality improvements than any prompting strategy. The optimal configuration uses citation instruction and agentic tools with no domain prompt, achieving 82.3% versus the 53.2% baseline (a 29.1pp improvement).

Beyond the main effects, the factor interactions reveal that factors do not act independently:

2. **Pair citation instruction with agentic tools.** Citation instruction is most effective when the model can actually retrieve and link real documentation. In agentic configurations, the Citation dimension jumps from 1.1/10 to 7.8/10. In non-agentic configurations, the model can only vaguely reference documentation without providing real URLs.

3. **Remove the domain prompt from agentic configurations.** A static knowledge primer shows a marginal negative main effect (−0.8pp) and actively interferes with agentic tool use in specific combinations. The best configuration (C+A) uses no domain prompt; adding the domain prompt (D+C+A) drops the score from 82.3% to 76.0%. The domain prompt is only marginally useful in non-agentic, single-call scenarios where the model has no other source of Snowflake-specific context.

4. **Do not add self-critique steps.** The generate-then-revise pattern degrades both score (−2.7pp) and must-have compliance (−9.8pp). This is the most consistent negative finding across the full 128-question dataset: self-critique hurts in every configuration where agentic tools are present.

For product teams configuring Snowflake AI developer tools, the prescription is straightforward: give the agent tool access, instruct it to cite sources, and stay out of its way.

Limitations

This benchmark has several limitations worth noting:

- **Single respondent model.** All 16 runs use `claude-opus-4-6`; results may differ for other models.
- **Question bank coverage.** The 128-question bank spans 32 categories with 4 questions each; individual category estimates carry higher variance than aggregate scores.
- **LLM-as-judge scoring.** Even with a 3-model panel, LLM judges may differ from human expert evaluation on nuanced questions. The must-have elements are binary checks that do not capture partial credit for closely related facts.
- **No TruLens integration in production scoring.** Although we built a TruLens integration (instrumented app, custom feedback functions, Snowflake connector), the 16-run factorial experiment used our custom 3-judge pipeline rather than TruLens. This means we lack standardized OpenTelemetry tracing of retrieval and generation spans, which would provide deeper observability into why agentic runs perform better. The custom pipeline also does not produce the RAG Triad metrics (groundedness, answer relevance, context relevance) that would enable direct comparison with other TruLens-evaluated systems. Migrating the scoring pipeline to TruLens would unify evaluation with Snowflake AI Observability and make results visible in Snowsight under AI & ML > Evaluations.

Next Steps

The immediate priorities are automating the pipeline for scheduled runs (to detect regressions as models and documentation change), expanding to multi-agent testing (Claude Code, Codex, Cursor) to understand competitive positioning, and building a PM-facing Streamlit interface for self-serve prompt configuration testing.

References

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